

A Study on Multisensory English Pronunciation Learning Support Based on Speech Prosody Visualization

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1. Introduction

With the development of mobile applications and online learning platforms, digital tools have been widely used in English pronunciation training. However, many existing systems rely on synthesized or relatively mechanical example audio with limited intonation variation. As a result, learners may imitate pronunciations that lack natural prosody. In addition, many approaches focus primarily on word-level practice, making it difficult for learners to master correct stress and intonation patterns when producing full sentences.

To address these issues, this study proposes a pronunciation learning support method based on speech prosody visualization. Acoustic features such as pitch and intensity are extracted from speech and mapped to visual variables including text size, color, and vertical position. Through different visualization designs and user testing, this study explores how visual representations can help non-native English learners more intuitively understand sentence-level prosody and improve pronunciation practice.

2. Problem

Speech prosody plays an important role in spoken English, reflected in variations of pitch, intensity, and stress. Previous studies note that prosodic features such as fundamental frequency and amplitude provide important cues for speech comprehension, conveys paralinguistic information that supports listeners' understanding of meaning and emotion.

However, non-native English learners often find it difficult to perceive prosodic variations relying only on auditory input. In natural speech, prosodic cues vary dynamically and can be difficult to detect. At the same time, many language learning applications rely on synthesized example audio with limited prosodic variation, and pronunciation practice is often conducted at the word level rather than the sentence level. These limitations make it difficult for learners to develop natural sentence-level pronunciation patterns.

3. Approach

According to Mayer's Cognitive Theory of Multimedia Learning (CTML), integrating verbal and visual information can significantly enhance learners' understanding and facilitate knowledge acquisition. Therefore, this study extracts acoustic features such as pitch and intensity from speech signals and maps them.

Speech Analysis

The OpenAI Whisper speech recognition model is used to transcribe the .wav audio file and obtain timestamp information for each word in the sentence. Based on the timestamp segmentation, the Librosa audio analysis library (Librosa 0.11.0) is used to extract acoustic features from each word segment. The average values of pitch (Hz) and intensity (dB-log energy) are calculated for each word.

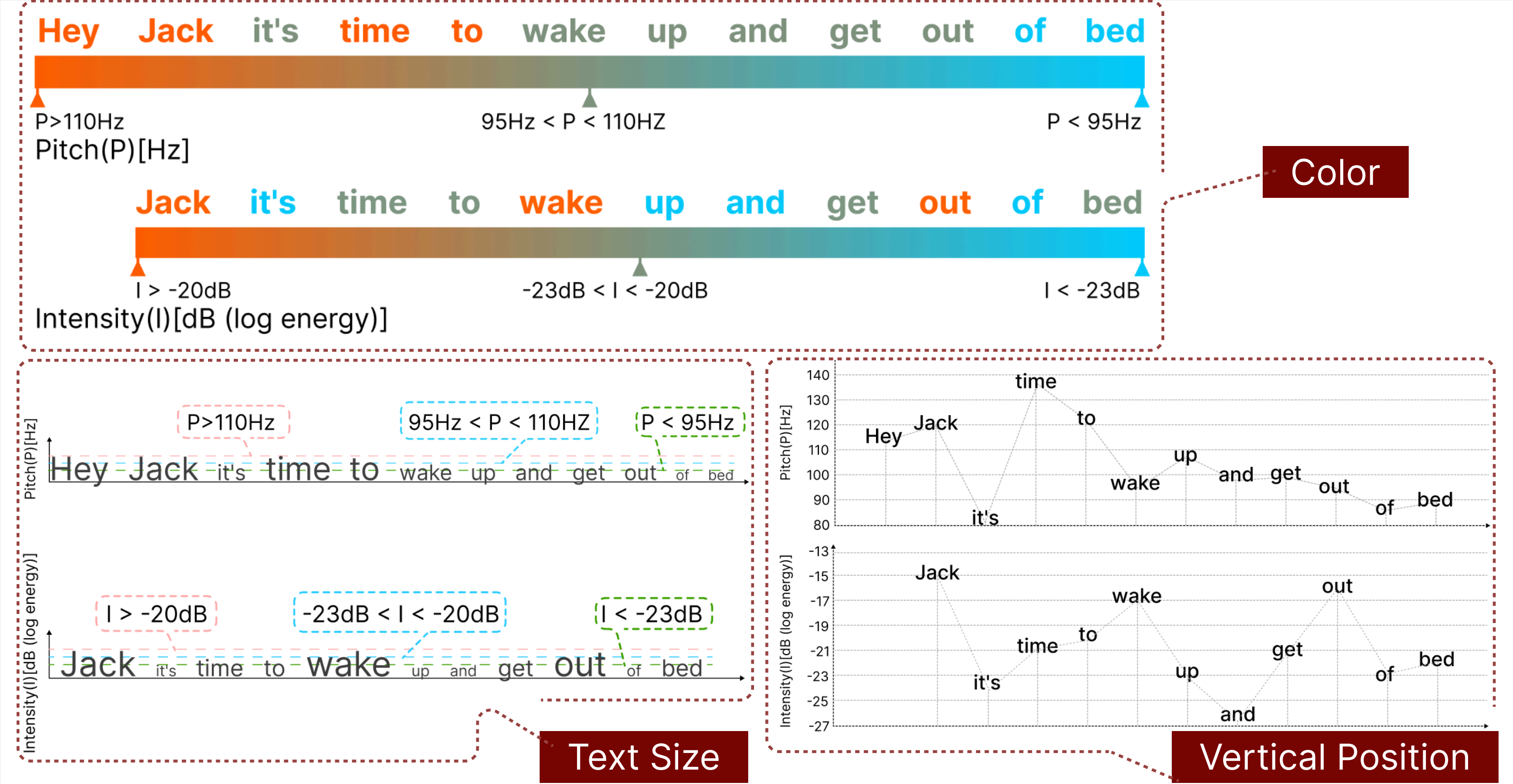
word	start	end	avg_pitch	avg_intensity
Hey,	0	0.86	113.851	-134.701
Jack,	1.36	1.66	120.477	-15.72
it's	2.14	2.42		-24.923
time	2.42	2.64	137.257	-21.003
to	2.64	2.86	121.351	-20.002
wake	2.86	3.08	95.013	-17.355
up	3.08	3.36	107.694	-23.072
and	3.36	3.52	97.366	-27.305
get	3.52	3.68	99.112	-21.294
out	3.68	3.86	95.607	-16.999
of	3.86	3.96	84.849	-23.521
bed.	3.96	4.2	88.913	-22.815

Data Processing

According to the overall numerical range of the extracted pitch and intensity data, the values are divided into three level intervals: low, medium, and high. These levels are used for subsequent visual variable mapping in order to enhance the visual distinction between different levels of speech features.

Visualization Design

The acoustic features after range classification are mapped to different visual variables, including text size, color, and vertical position. A total of six single-variable visualization schemes are designed.



User Evaluation

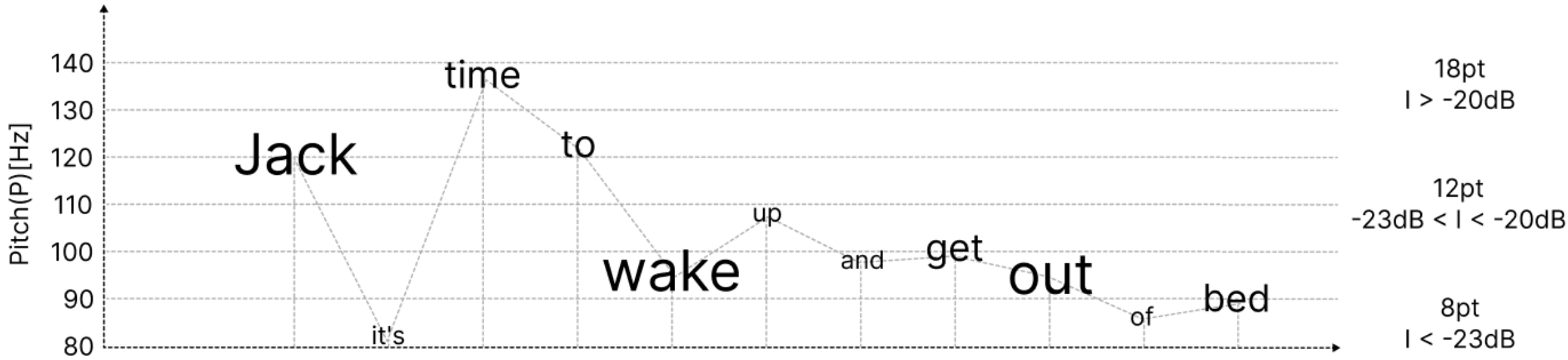
The user study is conducted through an online questionnaire. Participants first listen to the original sentence and then view different visualization representations. They evaluate each design based on readability and perceived usefulness for pronunciation learning.

4. Results

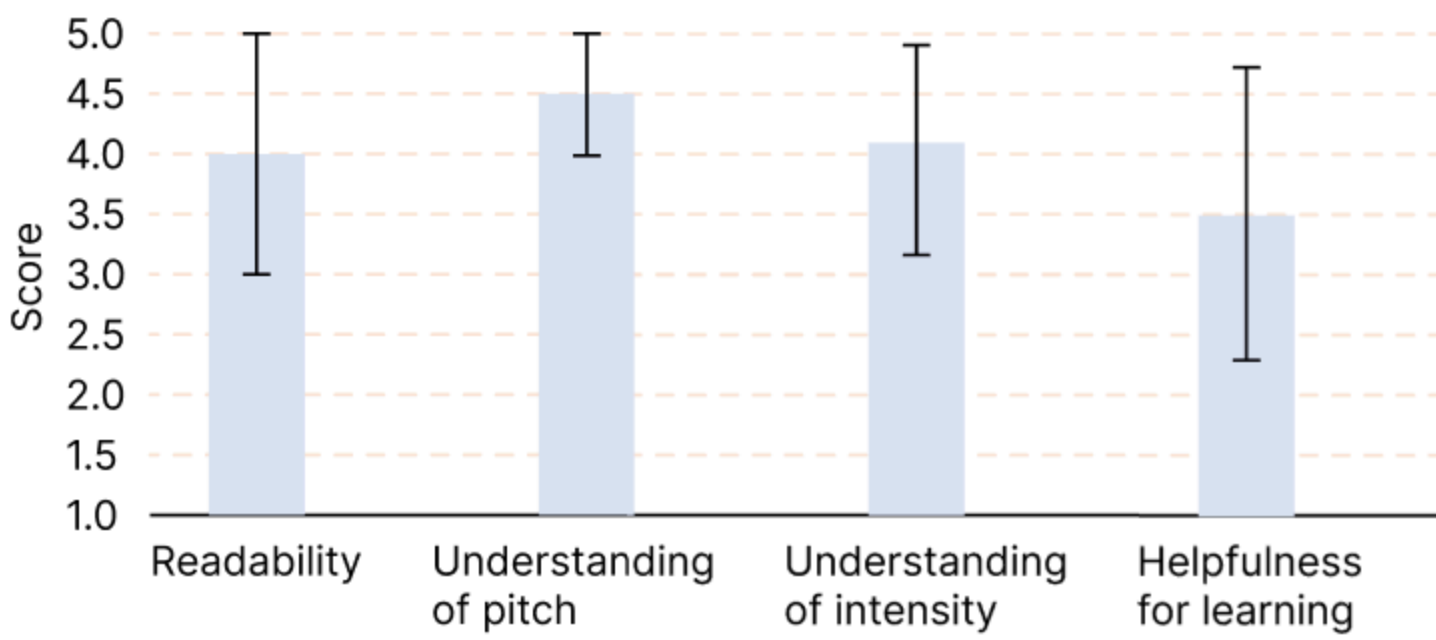
- Most participants preferred vertical text position for representing pitch. Text size was perceived as the most intuitive feature for representing speech intensity.

Feature	Method	Percentage
Pitch	Text Size	6.70%
	Color	6.70%
	Vertical Position	86.70%
Intensity	Text Size	66.70%
	Color	26.70%
	Vertical Position	6.70%

- Based on the preference results shown in the previous evaluation, a combined representation method was produced, in which vertical text position represents pitch and text size represents speech intensity.



- In order to evaluate the effectiveness of this visualization method, multiple aspects of the system were rated using a 5-point Likert scale. The related results were analyzed by calculating the Mean (M) and Standard Deviation (SD).



- As shown in figure, indicating that the proposed visualization method can effectively help users perceive and interpret prosodic features in speech, and the visualization method can present speech prosody in a clear and intuitive way, thereby supporting its potential as an assistive tool for non-native pronunciation learning.

Future work will focus on exploring the possibility of integrating this visualization method with gamified learning environments. One potential direction is to combine this visualization technology with mechanisms derived from 2D platform games, such as Mario-style jumping adventure games. This gamified interaction approach can serve as a guided learning tool. By embedding prosody training into interactive gameplay, it can help learners improve their perception and production of speech prosody in a more engaging and intuitive way.

5. References

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